The Natural Response of Uniform and Nonuniform Plates in Air and Partially Submerged in Water

S. Lee[1], E. Fishman[1], R. Pellegrini[2], A. Wang[1,3], K. T. Kiger[1], M. Yu[1], J. H. Duncan[1], F. Stern[4], M. Diez[2]

([1]Department of Mechanical Engineering, University of Maryland, College Park, [2]Institute of Marine Engineering, National Research Council, Italy, [3]Department of Civil, Environmental and Ocean Engineering, Stevens Institute of Technology, [4]Department of Mechanical Engineering, University of Iowa)

ABSTRACT

The free vibration of three aluminum plates (0.4 m wide, 1.08 m long) oriented horizontally is studied experimentally under two fluid conditions, one with the plate surround by air ("dry") and the other with the bottom plate surface in contact with a large undisturbed pool of water ("half-wet"). Measurements are made using cinematic Digital Image Correlation (DIC) of the top surface of the plate and optical tracking of the plate center point. Three plate geometries and boundary conditions are studied: a uniform plate with 6.35 mm thickness pinned at two opposite ends (UP), a uniform plate with 4.83 mm thickness simply supported at one end and clamped at the opposite end (UC) and a stepped plate with thickness varying from 12.7 mm to 6.35 mm pinned at two ends (SP). The plate's free response is induced using a large rubber mallet at a point slightly offset from the plate center point. Video frames of the plate's motion are collected from stereoscopic cameras and processed using LaVision StrainMaster and MATLAB to extract full-field displacements as a function of time. Two-degree-of-freedom displacements of the plate center are collected from tracking the center target's motion. Time and frequency response plots are presented for comparison between dry and half-wet conditions and analysis of their dynamics. It is found that the added mass of the water results in lower measured natural frequencies and increased damping. These results are compared to mode shapes and frequencies computed in COMSOL and found to agree.

INTRODUCTION

The high-speed impact of flexible structures on water surfaces, such as those found on high-speed planing boats or seaplanes, generate large impulsive loads that may generate a significant structural response. The interaction between the fluid and the structural deformation is transient and highly complex. A number of investigators have studied the impact of simplified structures, such as wedges and plates, on a water surface in order to explore fundamental aspects of the coupled dynamics. For the idealized impact of a flexible plate with a water surface, the impact begins with the plate's upper and lower surfaces surrounded by air and its trailing (lower) edge just touching the water surface and is considered completed when water reaches the plate's leading edge, leaving the upper surface dry and the lower surface in contact with water. This impact excites natural vibration modes of the plate and since the lower surface is in contact with water over at least some part of its length during the entire impact, the effect of the added mass of the water on just a single side of the plate is of interest.

The effect of added mass on vibrating structures immersed in a fluid has long been a subject of interest in the literature, arguably going back to the time of Bessel in the 19^{th} century (as relayed by Stokes (1850)). In more recent efforts, Yadykin et al. (2003) explored the influence of aspect ratio on single mode vibration of a cantilever plate submerged in water, finding that the added mass coefficients decreased with increasing modal frequency and decreasing plate aspect ratios. Meylan (1997) explored transforming the coupled fluid-structure equations into variational form and used a Rayleigh-Ritz method to develop numerical solutions of mode shapes of a floating plate. Chen et al. (2021) developed a numerical method and performed experiments with a partially submerged vertical plate, observing a decrease in natural frequency as the immersed portion of the plate increased. Using a novel numerical method, Liao and Ma (2016) investigated the frequency response characteristics of a plate with one side exposed to a compressible fluid. Kerboua et al. (2008) examined mathematically the vibration of simply supported and cantilevered rectangular plates fully submerged and located on the free surface of an incompressible irrotational flow. The vibration frequencies and added mass coefficients of the theory matched well with those from experiments and other theoretical models. While the literature briefly described above includes the results of experimental, theoretical, and numerical modal analysis for various fluid conditions and plate systems, the effects of complex plate boundary conditions, changes in plate structural uniformity and large-amplitude nonlinear vibrations have received little attention.

In our previous work (Pellegrini et al. (2020), Wang et al. (2022)), we have performed experiments and computational fluid dynamics (CFD) simulations for conditions of uniform aluminum plates during controlled constant velocity impact on a quiescent free surface. As a part of documenting the plate response characteristics, hammer-impact tests were conducted to determine the natural frequencies and damping coefficients of the plate/mount system under conditions where the plate is fully surrounded by air (referred to as "dry" conditions), and also where one side of the plate is in contact with the water surface (referred to as "half-wet" conditions). The plate response was monitored at 5 discrete points (3 along the longitudinal centerline and 2 on the lateral free edge) using a high-speed camera, which permitted determination of the frequency and damping ratio of the first order longitudinal mode and the first order transverse mode of the plate. The results showed that when the plate was placed with its lower surface in contact with the water the frequency of the first order longitudinal mode was reduced to approximately 1/3 of its dry-plate value. The frequency of the transverse mode, in contrast, was reduced to approximately 2/3 of its dry response.

The work described herein extends these prior rectangular plate studies to examine the influence of the mounting constraints, plate non uniformity and plate vibration amplitude on the natural frequencies, mode shapes and added masses. In these free vibration experiments, temporal records of the full field out-of-plane displacement of the central region of the plate's upper (dry) surface are measured using a cinematic Digital Image Correlation system. The local and average frequency response to a localized impulsive impact, along with their associated mode shapes are computed from the results. The tests are conducted for both the dry and half-wet conditions and the results are compared to obtain estimates of the added masses. Measurements are performed for three plate/mounting systems. In addition, fully coupled nonlinear numerical calculations of the plate natural vibration modes are performed and compared to the experimental results for one of the plate/mounting systems.

In the following, three subsections describing separately the experimental and computational details, the results and discussion and the conclusions are given sequentially.

EXPERIMENTAL & COMPUTATIONAL DETAILS

Plates and mounting systems

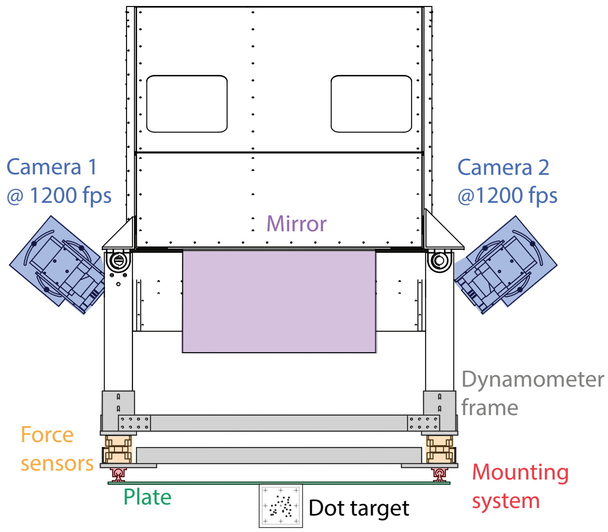
The plate dimensions, materials and mounting systems are described in this subsection. All three plates have the same lengths (L = 1079.5 mm) and widths (B = 406.4 mm). Two of the plates are made of aluminum 6061-T6 (density ρ = 2700 kgm^{-3}, elastic modulus E = 68.9 GPa, and Poisson's ratio ν = 0.33) and have uniform thicknesses, $h = 6.35$ mm (nominal) and 4.76 mm (nominal). The third plate is constructed from aluminum 5083 (ρ = 2660 kgm^{-3}, E = 71.0 GPa, and ν = 0.33) and its thickness varies over 10 equal steps from 12.7 mm (nominal) at the leading edge to 6.35 mm (nominal) at the trailing edge.

Following the procedures outlined in Wang et al. (2022), the plate thicknesses were measured with an ultrasonic thickness gauge (GE CL5 Krautkramer, with Alpha2-DFR probe) at locations on a 7-by-6 grid on the uniform thickness plates and a 5-by-2 grid on each thickness of the stepped plate. The measurement resolution of the ultrasonic meter is 0.001 mm. The measured average thicknesses of the uniform plates were $h = 6.431 \pm 0.008$ mm and 4.806 ± 0.009 mm and the thicknesses of the ten steps of the third plate are given in table 1.

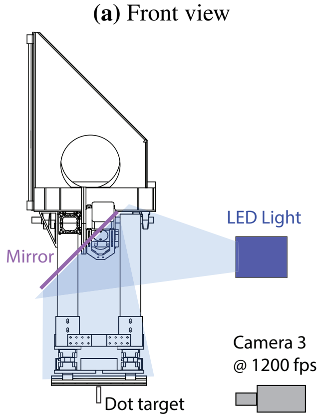
Table 1: Measured average thicknesses and standard deviations of the 10 steps of the stepped plate.

Step	h (mm)	σ (mm)
1	12.723	0.006
2	11.987	0.013
3	11.274	0.008
4	10.577	0.005
5	9.814	0.019
6	9.044	0.025
7	8.330	0.014
8	7.664	0.012
9	7.011	0.012
10	6.292	0.012

The experiments are conducted in a 13.4 m long, 2.45 m wide, and 1.33 m tall towing tank constructed with a steel frame and 35 mm-thick clear acrylic panels. A two-axis (horizontal and vertical motion) carriage system is mounted on a separated steel frame located along one of the long side walls of the tank. This carriage system was used for our previous plate-slamming experiments, see for



(a) Front view



(b) Side view

Figure 1: A schematic diagram of the carriage, aluminum plate, force sensors and stereo DIC system from the front and side is shown.

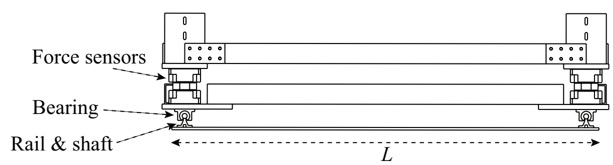
example Wang et al. (2022), and is depicted in figure 1 of the present paper.

In the present experiments, the carriage remains stationary with the plate held horizontal from the mounting points at its leading and trailing edges (zero pitch and roll angles), see figure 1. The carriage includes a four-sensor dynamometer to which the plate mounts are directly attached at the longitudinal ends of the plate.

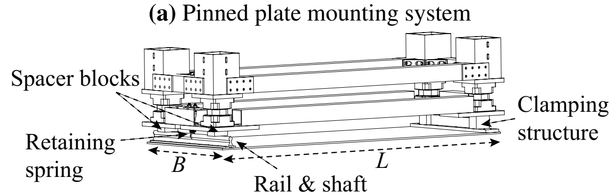
The thickest uniform plate, $h = 6.43$ mm, and the stepped plate are mounted to the dynamometer via identical pinned supports at the plate's leading and trailing edges. These are the supports used in Wang et al. (2022) and they allow the leading and trailing edges of the plate to have free rotation about transverse horizontal axes located 32 mm inward of the plate's short edges and 28.6 mm above the plate's upper surface. More details about the pinned supports can be found in Wang et al. (2022) and the schematic drawing given in figure 2(a).

The second uniform plate ($h = 4.806$ mm) is mounted to the dynamometer via a clamped leading edge and a simple support at the trailing edge. These mounts provide an approximate fixed mounting condition at the leading edge (no rotation or translation) and a free translation and rotation at the trailing edge. The clamping structure is 38.1 mm wide and its length extends to the full width of the plate. The simple support structure utilizes the rail and shaft structure from the pinned connection and is held in contact with the lower surface of aluminum spacer blocks by a retaining spring, see figure 2(b).

In the following, the plate with $h = 6.43$ mm is called the uniform pinned (UP) plate, the plate with $h = 4.81$ mm is called the uniform clamped (UC) plate and the plate with a stepped profile is called the stepped pinned (SP) plate.



(a) Pinned plate mounting system

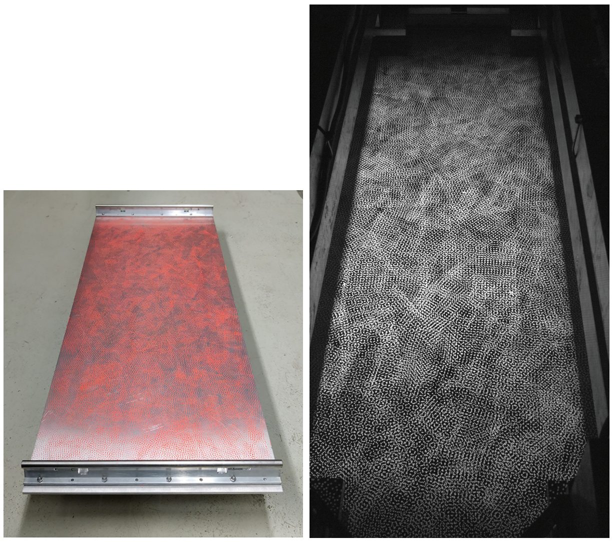


(b) Clamped plate mounting system

Figure 2: Schematic diagrams of the pinned and clamped plate mounting systems are shown in (a) and (b), respectively.

Plate deflection measurements

The temporal histories of the out-of-plane deflection of the plates are measured using cinematic digital image correlation (DIC) and a target tracking system. For the DIC measurements, the top side of each plate is detailed with a speckle pattern consisting of fluorescent paint applied using a custom paint roller as shown in figure 3(a). Stereo movies of the plate motion are taken with two Phantom VEO4K 990L cameras (8-mega pixel sensors) with each camera mounted on a separate beam that crosses the width of the tank and clamped to the steel tank frame near the leading and trailing edges of the plate. This mounting system isolates the cameras from the plate motion. The cameras are placed approximately 800 mm from the plate center and rotated such that the optical axis of the 24 mm lens



(a) **(b)**

Figure 3: In (a), an example image of the speckle pattern applied to the plate is shown. In (b), an example image of the speckle pattern applied to the plate as seen by the left camera through the filter is shown.

is 40 degrees below horizontal and intersects with the center of the plate. A Scheimpflug adapter is used to mount the lens, ensuring the focal plane is parallel to the static, undeformed surface of the plate. Each lens is outfitted with a 530 nm long-pass filter to block light wavelengths emitted by the LED array described below while passing light emitted by the fluorescent paint. The cameras capture gray-scale images at a resolution of 4096 pixel x 1728 pixel images simultaneously at 1,200 frames/sec, with a maximum capture time of 5.5 sec. Due to line-of-sight blockage by the carriage components, the DIC system records the motion only in a rectangular region of the plate as shown in figure 4. The length (L_D) and width (B_D) of the DIC measurement window on the plate surface are approximately 700 mm and 250 mm, respectively.

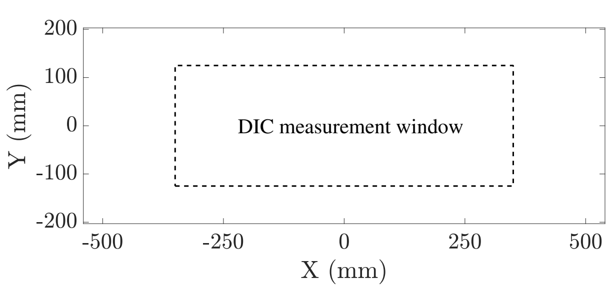


Figure 4: The DIC measurement window ($L_D \approx 700$ mm and $B_D \approx 250$ mm) is shown in the dashed rectangle relative to the plate where the bounds of the plot are the plate dimensions ($L = 1079.5$ mm and $B = 406.4$ mm).

The plate is illuminated by an LED Array Module (LaVision LED-300) that is mounted to the tank's steel frame, off to the side and above the level of the plate. An acrylic mirror mounted to the carriage redirects light from this array downward onto the plate surface. The timing synchronization of the cameras and LED array is controlled by a LaVision PTU-X. An example image of a plate as seen by the camera is shown in figure 3(b). Each stereo image pair is processed using LaVision StrainMaster to obtain maps of the plate out-of-plane deflection on a grid of approximately 160-by-50 data points with a grid spacing of 5.5 mm x 5.5 mm. The mode shapes are obtained from a single trial's time history of plate out-of-plane deflection through the following steps:

1. Compute the frequency response of the deflection time history of each data point using Fast Fourier Transforms (FFTs).

2. Average the frequency response across all data points and identify resonant frequencies of the measured area of the plate.

3. Generate mode shapes of resonant frequencies by plotting the amplitude at the respective frequency at each data point.

For the target tracking measurements, a small target (approximately 2 cm by 2 cm) consisting of a white surface covered with black dots was affixed to a 3D-printed L-shaped bracket. This bracket was glued to the center of the plate's bottom surface such that the target image was perpendicular to the plate, see figure 1. A third camera (Phantom V640 (4-mega pixel sensor)) mounted on a tripod outside the tank records the motion of the target. The target is tracked in the recorded image sequence using the target tracking analysis methods described in Wang et al. (2022) to obtain a time history of out-of-plane deflection at the plate center.

Experimental procedures

Each experimental run begins by striking the plate with an impact hammer (PCB Piezotronics Model 086C01) or a large rubber mallet. The cameras and the light operate continuously before the hammer impact, though images are not permanently saved. The cameras are set to pre-trigger the impact in order to capture a non-deformed plate shape as well as the impact dynamics. Piezoelectric voltage readings from the hammer are recorded using an N-Scope USB oscilloscope. In this paper, only results from strikes with the large rubber mallet are presented. The experiment concludes after the camera memory is filled, totaling in approximately 7,000 frames recorded from each camera. Video frames from the two DIC cameras are processed using LaVision StrainMaster to produce a time history of out-of-plane deflections. The plate is struck on the bottom surface for "dry" impact conditions and on the top surface for the "half-wet" conditions. In the half-wet conditions, each plate is submerged to approximately half its thickness and a high-speed camera placed underneath the tank captures the lower surface of the plate to ensure no air pockets are trapped between the water surface and the plate's lower surface during the experiment. For the stepped pinned plate, the upper surface is aligned to horizontal so the submergence depth increases as the plate thickness increases. In all cases, the plate is struck near the plate center offset by about 2 inches in both the transverse and longitudinal directions to avoid nodal lines. Each case is repeated 3 times for a total of 18 distinct trials across all plates.

A plot of deflection versus time at the center of the plate from a uniform pinned plate trial with curves from both the DIC and target tracking measurements is given in figure 5. The two curves are nearly identical and show decaying oscillations at primarily two widely separated frequencies.

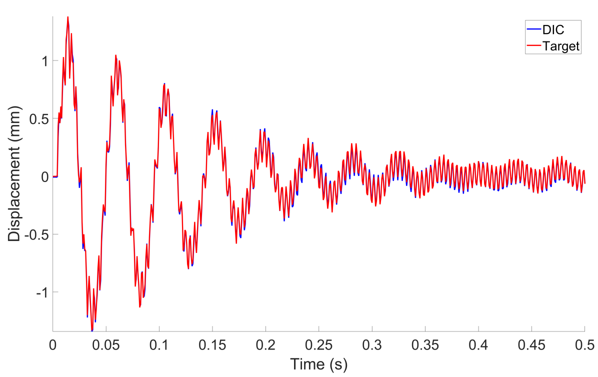


Figure 5: Comparison of the raw displacement measured by DIC and the deflection target for the uniform pinned plate.

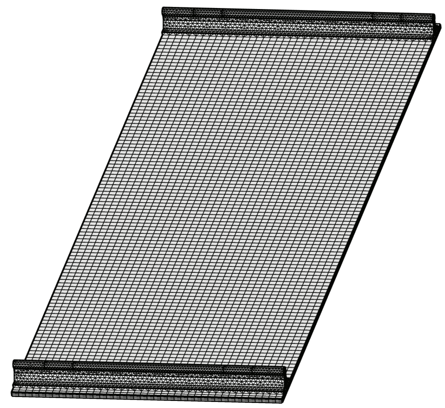


Figure 6: The FE mesh of the plate and rails are meshed with about 66,000 elements.

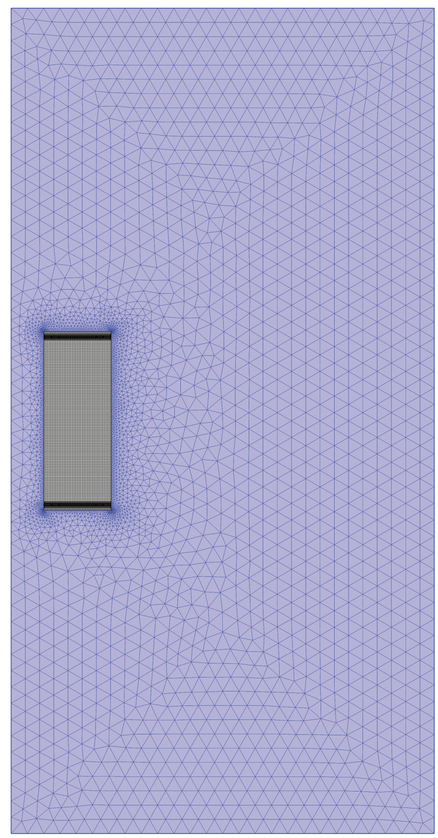
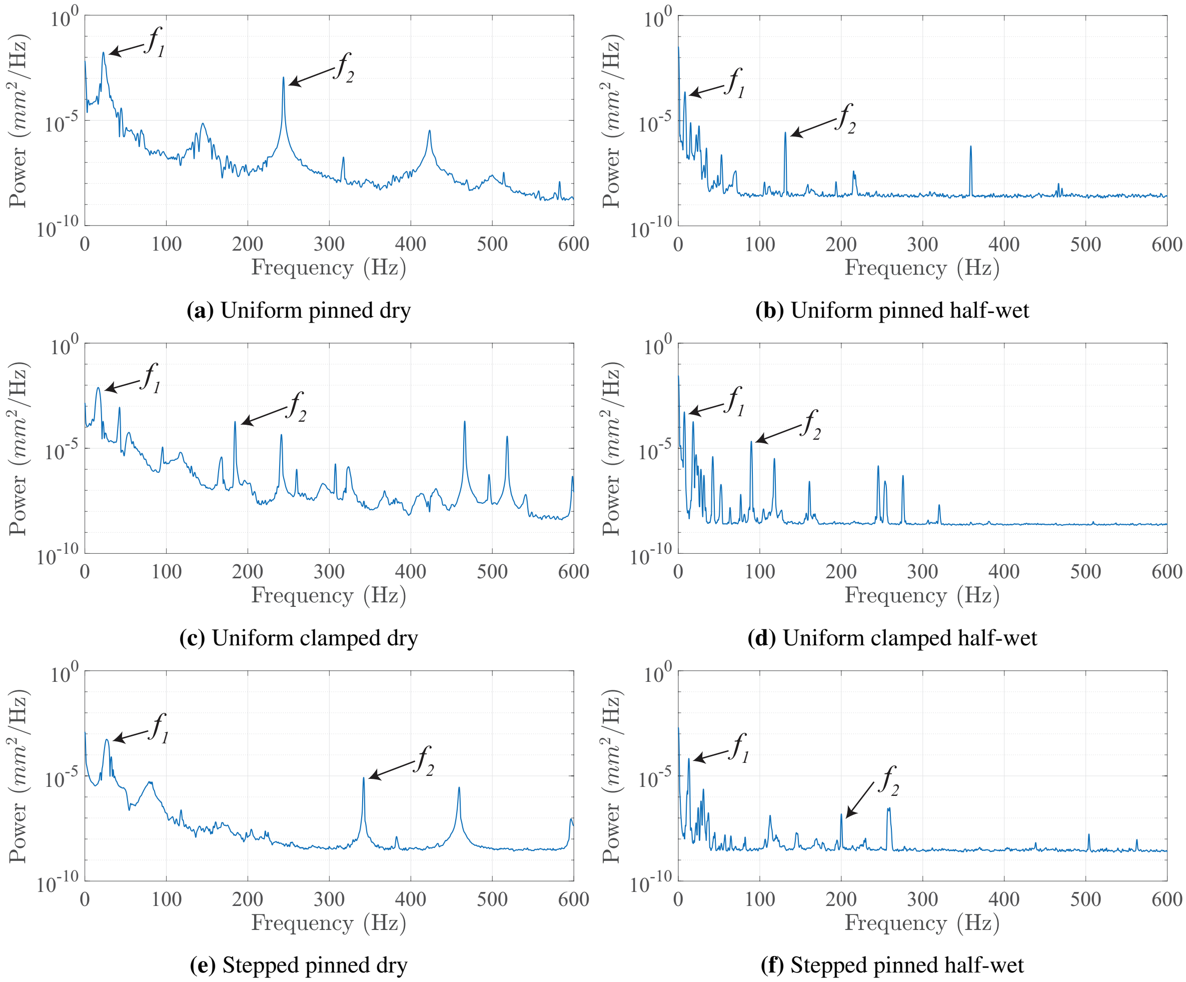


Figure 7: The FE mesh of the surrounding volume for the half-wet analysis is meshed with about 325k elements elements.

Computational Methods

Numerical studies focus on (a) the eigenfrequency analysis in dry and half-wet conditions and (b) transient analysis in half-wet conditions subject to impulse excitation. For the eigenfrequency analysis, the finite element (FE) code COMSOL Multiphysics® is used, where structural elements are coupled with acoustic elements to identify half-wet modes and frequencies.

The meshed FE model of the 6.61-mm-thick aluminum plate and mounting rails is shown in figure 6. The mesh consists of about 66,000 elements. Spring foundation boundary conditions with an elastic constant of 1.05E9 N/m are used for the rails (pins) that



(a) Uniform pinned dry

(b) Uniform pinned half-wet

(c) Uniform clamped dry

(d) Uniform clamped half-wet

(e) Stepped pinned dry

(f) Stepped pinned half-wet

Figure 8: Experimentally measured deflection power spectrum averaged of the surface of the plate for the six experimental conditions. The frequency peaks chosen for f_1 and f_2 are noted in each plot.

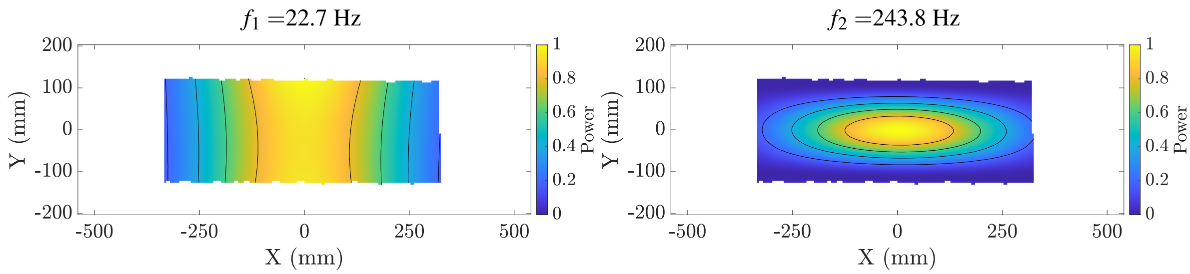
connect the mounting system with the carriage in the experiments. The value of the elastic constant is identified by matching the first eigenfrequency and minimizing the strain prediction error at the center of the plate during vertical slam between simulations and experiments (Pellegrini et al. (2020)). Figure 7 shows the mesh of the volume surrounding the plate for the half-wet eigenfrequency analysis. The volume is meshed with about 325k elements, and has the same size in width and height as the experimental facility described above.

Table 2: The frequencies of modes 1 and 2 from experiments and numerical calculations, and the damping ratios of mode 1 from experiments.

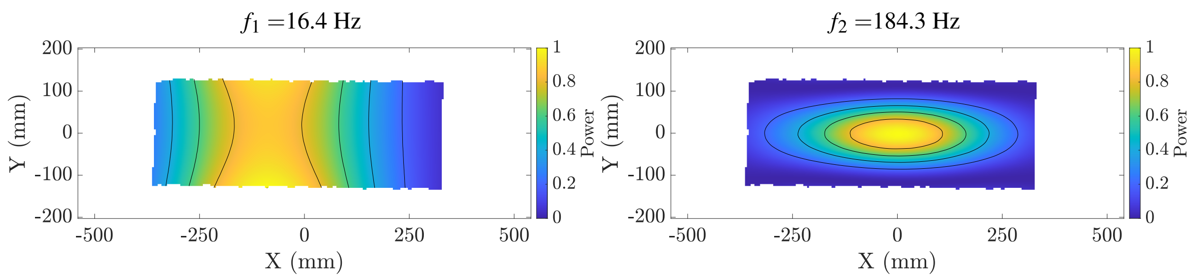
Plate	Condition	f_1 [Hz]	f_2 [Hz]	ζ_1	ζ_2
UP_{exp}	Dry	22.7	243.8	0.0409	0.0009
UP_{exp}	Half-wet	8.2	131.4	0.0621	0.0014
UP_{num}	Dry	24.3	250.3		
UP_{num}	Half-wet	7.93	136.1		
UC_{exp}	Dry	16.4	184.3	0.0860	0.0007
UC_{exp}	Half-wet	7.5	89.7	0.1185	0.0019
SP_{exp}	Dry	26.7	342.1	0.0632	0.0009
SP_{exp}	Half-wet	12.9	200.0	0.0909	0.0012

RESULTS & DISCUSSION

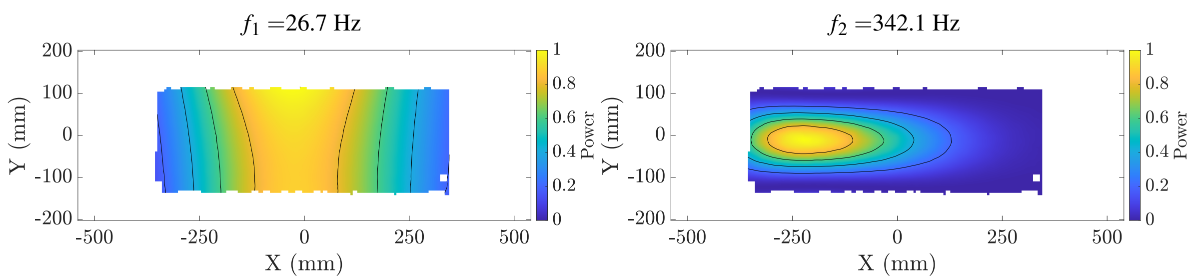
The out-of-plane deflection data from the DIC measurements is processed to obtain the natural frequencies, damping ratios and mode shapes for the three plates in the dry and half-wet conditions. The spatially averaged spectra shown in figure 8 reveal numerous large-amplitude peaks. Two modes identified from previous experiments (Wang et al. (2022)) were a longitudinal first order mode at a low frequency and a transverse first order mode at a higher frequency. The



(a) Uniform pinned dry



(b) Uniform clamped dry



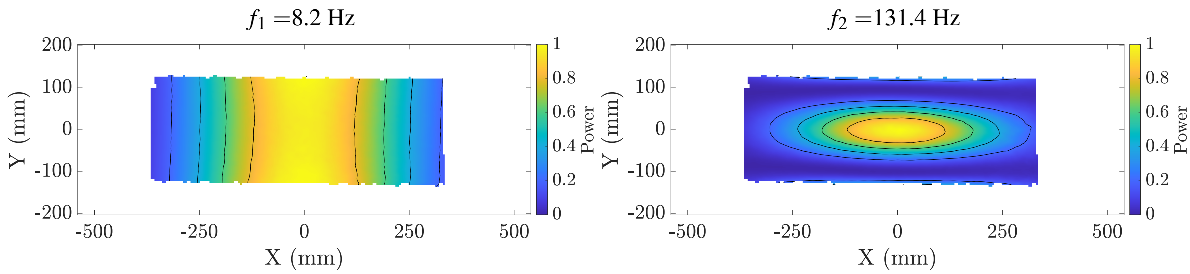
(c) Stepped pinned dry

Figure 9: Normalized plate vibration mode shapes obtained experimentally are shown for the three plates in the dry condition. For the uniform clamped plate, the clamped boundary condition is on the right hand side of the plot. For the stepped pinned plate, the thickest section of the stepped plate is at the right hand side.

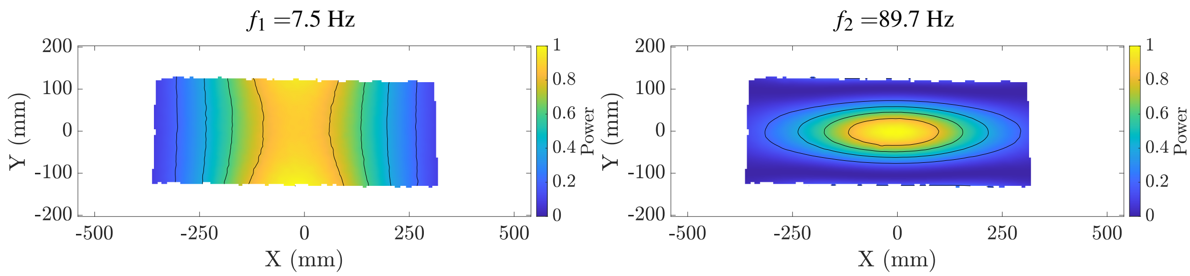
frequencies associated with these two modes are identified for each plate and condition where the lower frequency is labeled f_1 and the higher frequency is labeled f_2 (see table 2). For all plates, the power levels of the peaks in the dry condition are at least one order of magnitude larger than the power levels of the peaks in the half-wet condition indicating that the plate oscillation amplitude is much larger in the dry condition. In all cases, the f_1 mode is the dominant mode as the power level of f_1 is at least one magnitude larger than the power level of f_2.

The normalized mode shapes associated with frequencies f_1 and f_2 are given in figure 9 for the dry plate conditions and figure 10 for the half-wet plate conditions. As can be seen from the three plots on the left in figure 9, f_1 corresponds to a longitudinal first order bending mode with a transverse axis near the center for all three plates. For the uniform plate with pinned supports (UP), the maximum deflection is very close to the plate mid point ($X = 0$ mm). For the uniform plate with clamped/simple supports (UC), the maximum of the mode shape moves toward the left (simply-supported end), likely due to the ease in both rotation and translation provided by the simple support. For the stepped plate with pinned supports (SP), there is only a slight shift of the maximum towards the left (thinner end) due to the plate's bending stiffness decreasing with each step as the plate thins from right to left.

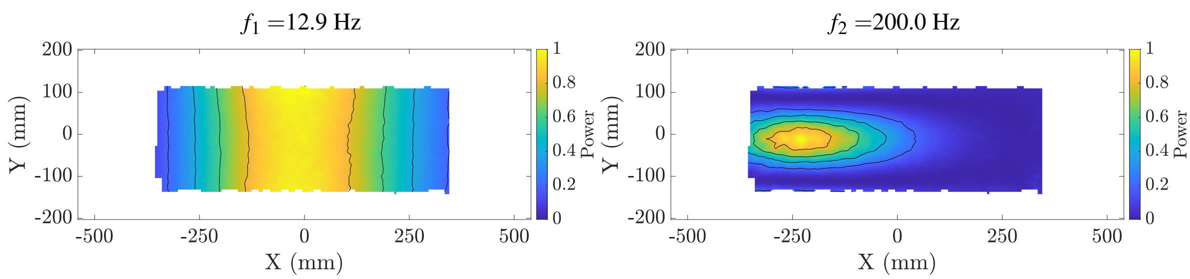
The f_2 mode shapes are shown in the three plots on the right in figure 9. Unlike in the f_1 case, the f_2 mode position is not noticeably biased from the plate center ($X = 0$ mm) for either the UP or UC plates; however, for the SP plate, there is a strong bias towards the left hand side of the plate. This is due to the increased flexibility of this plate as the thicknesses decreases towards the left.



(a) Uniform pinned half-wet



(b) Uniform clamped half-wet



(c) Stepped pinned half-wet

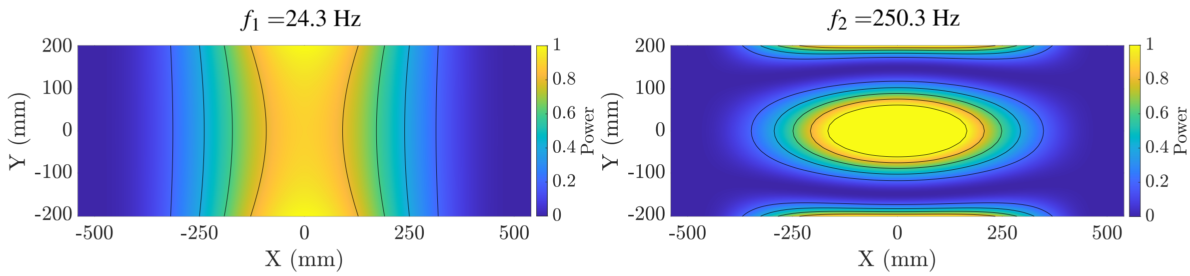
Figure 10: Normalized plate vibration mode shapes obtained experimentally are shown for the three plates in the half-wet condition. For the uniform clamped plate, the clamped boundary condition is on the right hand side of the plot. For the stepped pinned plate, the thickest section of the stepped plate is at the right hand side.

The half-wet mode shapes as shown in figure 10, have qualitatively the same shapes as the dry mode shapes in figure 9. This is an indication that the frequencies f_1 and f_2 in the dry and half-wet cases correspond to the same modes and that the changes in frequency between the dry and half-wet cases can be used to explore the added mass for each plate. The frequencies and mode shapes from the numerical calculations for the UP plate compare well to results obtained experimentally (see table 2 and figure 11, respectively).

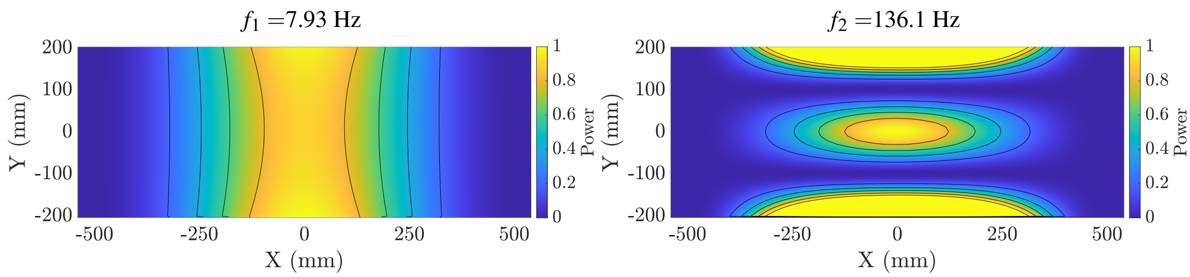
The damping ratio corresponding to each f is obtained from the out-of-plane deflection time history at the center of the plate. This signal is pre-processed by band-pass filtering at each frequency f. The solution of linear damping equation,

$$z(t) = Ae^{-\zeta\omega t}\sin\left(\left(1-\zeta^2\right)^{0.5}\omega t+\phi\right), \qquad (1)$$

is fit to the signal to obtain numerical values of the damping ratio ζ. However, the oscillation frequency and damping ratio both shift as the amplitude decreases from oscillations on the order of 1 mm to 0.1 mm. This shift is a nonlinear effect attributed to friction in mounting systems and asymmetries in the plate geometry. The solution of the linear damping equation is fit to the beginning of the signal when the amplitude is still large on the order of 1 mm and the response is linear. The results for all cases are listed in table 2. The damping ratios for all plates at f_1 are on the same order in both dry and half-wet conditions. The same is true for the damping ratios for all plates at f_2. This indicates that the free vibration is dominated by structural damping. For both f_1 and f_2, the damping ratios increase from dry to half-wet due to additional damping provided by water in contact with the lower surface of the plates.



(a) Uniform pinned dry



(b) Uniform pinned half-wet

Figure 11: Uniform pinned dry and half-wet mode shapes computed in COMSOL.

The frequency ratios f_2/f_1, f_{1HW}/f_{1D} and f_{2HW}/f_{2D} are given in table 3. The mode 2 frequencies are on the order of 10 times higher than the mode 1 frequencies and the ratio f_2/f_1 is higher in half-wet cases than in the dry cases. The ratios of the half-wet to dry frequencies for mode 1 range from 0.33 to 0.48 and are larger for mode 2, ranging from 0.49 to 0.58. The half-wet to dry frequency ratios can be used to compute the ratios of the added mass to the plate mass for the two modes for each plate by the following method. The generalized formula for the ratio of dry to half-wet frequencies of any given mode is,

$$\frac{f_{\text{dry}}}{f_{\text{half-wet}}} = \frac{\left[\sqrt{\frac{k}{m_p+m'}}\right]_{\text{dry}}}{\left[\sqrt{\frac{k}{m_p+m'}}\right]_{\text{half-wet}}} \tag{2}$$

where m_p is the plate mass, m' is the added mass and k is the plate generalized stiffness. Assuming that k is independent of the plate submergence and $m' = 0$ for the dry case, the added mass ratio $M = m'/m_p$ can be found. The values of M for modes 1 and 2 for each plate are presented in table 4. In all cases, the added mass has a larger effect on mode 1 compared with the effect on mode 2. For the UP plate, the added mass ratio of mode 1 is about three times the added mass ratio of mode 2. For the UC plate, the added mass ratio of mode 1 is similar to the added mass ratio of mode 2. For the SP plate, the added mass ratio of mode 1 is about one and a half times the added mass ratio of mode 2.

Table 3: Frequency ratios from experimental measurements and numerical calculations.

Plate	Condition	f_2/f_1	f_{1HW}/f_{1D}	f_{2HW}/f_{2D}
UP_{exp}	Dry	10.74		
UP_{exp}	Half-wet	16.02	0.36	0.54
UP_{num}	Dry	10.30		
UP_{num}	Half-wet	17.16	0.33	0.54
UC_{exp}	Dry	11.24		
UC_{exp}	Half-wet	11.96	0.46	0.49
SP_{exp}	Dry	12.81		
SP_{exp}	Half-wet	15.50	0.48	0.58

Table 4: Added Mass Ratios $M = \frac{m'}{m_p}$

Plate	M_1	M_2
UP_{exp}	6.7	2.44
UP_{num}	8.4	2.38
UC_{exp}	3.8	3.22
SP_{exp}	3.2	1.93

CONCLUSION

The natural response of three different plates (uniform pinned, uniform clamped and stepped pinned) of varying thickness and mounting conditions is measured and computed for a dry condition and a half-wet condition where the plate's lower surface is completely in contact with water. The plate motion is measured with cinematic DIC and a target tracking method to obtain a time history of the out-of-plane deflection, from which spectra can be computed. The natural frequencies and corresponding mode shapes for the uniform pinned plate in both the dry and half-wet conditions are also computed using COMSOL. For each plate and condition, two frequency peaks of interest are identified where the lower frequency f_1 is a first order longitudinal bending mode and the higher frequency f_2 is a first order transverse bending mode. For the uniform clamped and stepped pinned plates, the location of maximum oscillation in some modes (UC mode 1, SP mode 1, SP mode 2) shifts towards the less stiff portion of the plate (simply supported edge for UC and thinner portion for SP). For all plates, the power level of the signal decreases by about one order of magnitude from the dry condition to the half-wet condition due to the added mass. For all cases, the power level of f_1 is typically one order of magnitude greater than the power level of f_2. The damping ratios at f_1 and f_2 are computed for each plate in dry and half-wet conditions. The damping in the system is primarily structural damping and the damping increases in the half-wet condition. The frequencies of f_1 and f_2 in the dry and half-wet cases are used to compute an added mass ratio M' that indicates the effect of added mass on that particular mode. The added mass effect is greater on mode 1 than on mode 2 for all plates. The frequencies, mode shapes, and added mass ratios compare well between results for the uniform pinned plate obtained experimentally and computed with COMSOL.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge the support of the Office of Naval Research, Ship Systems and Engineering Research Division (Code 331), Program Manager Dr. Robert Brizzolara: University of Maryland, grant N00014-20-1-2349; University of Iowa and the Institute of Marine Engineering, grant N00014-20-1-2259.

REFERENCES

Chen, G.-W., Liao, C.-Y., Lin, Y.-Z., and Ma, C.-C., "Analytic solution to the coupled vibration characteristics of a rectangular plate partially immersed in a finite fluid container," Journal of Sound and Vibration, Vol. 515, no. 116446, 2021, pp. 1–32.

Kerboua, Y., Lakis, A., Thomas, M., and Marcouiller, L., "Vibration analysis of rectangular plates coupled with fluid," Applied Mathematical Modelling, Vol. 32, 2008, pp. 2570–2586.

Liao, C.-Y. and Ma, C.-C., "Vibration characteristics of rectangular plate in compressible inviscid fluid," Journal of Sound and Vibration, Vol. 362, 2016, pp. 228–251.

Meylan, M. H., "The forced vibration of a thin plate floating on an infinite liquid," Journal of Sound and Vibration, Vol. 205, no. 5, 1997, pp. 581–591.

Pellegrini, R., Diez, M., Wang, Z., Stern, F., Wang, A., Wong, Z., Yu, M., Kiger, K. T., and Duncan, J. H., "High-fidelity fsi simulations and v&v of vertical and oblique flexible plate slamming," Proceedings of the 33rd Symposium on Naval Hydrodynamics, 2020.

Stokes, S. G. G., "On the effect of the internal friction of fluids on the motion of pendulums," Transactions of the Cambridge Philosophical Society, Vol. 9, 1850, pp. 8–106.

Wang, A., Wong, K. P., Yu, M., Kiger, K. T., and Duncan, J. H., "The controlled impact of elastic plates on a quiescent water surface," Journal of Fluid Mechanics, Vol. 939, no. A4, 2022, pp. 1–43.

Yadykin, Y., Tenetov, V., and Levin, D., "The added mass of a flexible plate oscillating in a fluid," Journal of Fluids and Structures, Vol. 17, no. 1, 2003, pp. 115–123.

DISCUSSION

Kelli Hendrickson
University of North Carolina at Chapel Hill

The authors present initial results in understanding the change of the natural response of uniform and nonuniform plates when partially submerged in water using experiments and simulations. The manuscript mainly details the experimental results for the linear dynamic response of the plates in varying pinned-clamped configurations as well as for one plate with non-uniform thickness. The authors observe a downward shift in the natural frequencies and modified mode shapes due to the presence of the water on one side of the plate.

1. Please comment on any effect (expected or observed) of the plate submergence depth on the half-wet results. For the plate with a stepped thickness, what is the variation of the plate submergence along the plate?

2. Do the authors observe any measurable water surface response during the experiments and simulations?

3. In the uniformed pinned half-wet condition vibration modes (Figure 10a), there appears to be an angle to the shape of the mode near x=0 compared to that of the uniform clamped and stepped pinned results. Is this consistent in all of the trials and if so what might be the reason for this?

AUTHOR'S REPLY

Thank you for your comments and questions, each of which is addressed below.

Question 1: The effect of plate submergence depth should be negligible since it adds only a hydrostatic pressure to the bottom of the plate. Also, the plate motion is perpendicular to the normal to the plate edge surfaces, further decreasing any added mass effect from the plate edge. For the stepped pinned plate, the upper surface of the plate along the incline is aligned to horizontal so the plate submergence increases as the plate thickness increases.

Question 2: During the experiments, we have observed that small water surface ripples are emitted from the plate edges during the free plate vibration. The generation of these waves by the plate vibrations is inefficient since the plate vibration wavelengths, which are on the order B and L, are much larger than the wavelengths of the gravity capillary waves and capillary waves with the same frequencies as the vibration modes, which are ≥ 8 Hz. Thus, we believe that this effect will not significantly affect the plate vibrations.

Question 3: The angle in the shape of the half-wet lowest frequency mode plot did not appear consistently. In later experiments, vibrations with larger initial amplitude were studied. In these new experiments, the broad lowest mode peak shown in the spectrum in figure 8(a), becomes sharp, the noise level in the mode shape at this frequency drops dramatically, and the shape becomes symmetric about the x-y axes in the plot. We believe that this effect is associated with the damping of the system, which causes the lowest mode to disappear in few oscillation cycles when its initial amplitude is small.

DISCUSSION

Lian Shen
University of Minnesota

The paper reports an interesting and valuable experimental and numerical study of the vibration of a plate induced by an impact hammer. A dry condition, with the plate surrounded by air on both sides of the plate, and a half-wet condition, with one side of the plate in contact with water, are compared to investigate the added mass effect. My question concerns the project's motivation, high-speed planing boats and seaplanes. Could the authors elaborate on how the dominant frequencies and the associated mode shapes identified in this study may guide the naval applications and what studies are needed in the next step of research?

AUTHOR'S REPLY

Thank you for your comments and question, which is addressed below.

As the plate submerges, transitioning from dry to half-wet, the effect of added mass becomes increasingly larger. The quantified added mass ratio is directly correlated to plate flexibility and can help inform the expected behavior (e.g., force vs time record and spray root propagation vs time) for a plate design. Future studies should focus on observing bending modes during the impacts, in particular the extreme short duration impacts that can occur when portions of the surface of the plate are nearly parallel to the local water surface. This scenario can occur for plate impact on a wavy free surface.